

Solving Combined Field Integral Equation With Deep Neural Network for 2-D Conducting Object

Rui Guo , Graduate Student Member, IEEE, Zhichao Lin , Tao Shan , Maokun Li , Senior Member, IEEE, Fan Yang , Fellow, IEEE, Shenheng Xu , Member, IEEE, and Aria Abubakar , Senior Member, IEEE

Abstract—Solving the combined field integral equation (CFIE) for the large-scale scattering problem is computationally expensive. In this letter, we investigate the feasibility of applying deep learning to solve the CFIE for 2-D perfect electrically conducting objects. Inspired by the conjugate gradient method, an iterative deep neural network is designed to learn the manner of solving the surface current density from the CFIE, with the input being the coefficient matrix of the equation. This process involves physics through surface integration and need less iterations than the conventional iterative equation solver. In numerical tests, we evaluate the network's performance by comparing the predicted surface current density and bistatic scattering cross section with the solutions rigorously computed. This method provides an insight into applying machine learning techniques together with electromagnetic (EM) physics to fast EM computation with the same level of accuracy as traditional method.

Index Terms—Combined field integral equation (CFIE), deep learning, method of moments (MoM), physics embedded, scattering problem.

I. INTRODUCTION

SURFACE integral equations are well suited to solve the scattering problem for perfect electrically conducting (PEC) objects. They only require discretization of the object surface, resulting in less unknowns than volume integral equations [1], [2]. Both electric field integral equation (EFIE) and magnetic field integral equation (MFIE) are applicable to solve the discrete surface current density. But EFIEs and MFIEs can be singular at the frequencies corresponding to the resonant frequencies of a cavity formed by perfect conductors [3]. This problem can be eliminated by constructing the combined field integral equation (CFIE). After applying the method of moments (MoM) [3], the

equation converts to a matrix equation, thereby allowing us to obtain the solution of the CFIE by solving the matrix equation.

For large-scale scattering problems, the processes of both establishing and solving the matrix equation can be time consuming. To accelerate numerical computation, various methods have been developed, such as the fast Fourier transform method [3], the multilevel fast multipole algorithm [4], and the reduced basis method [5], [6]. Especially, in [5], the authors proposed a two-step approach for fast computation, where the reduced basis is constructed offline by computationally intensive process but the surface current is generated by an inexpensive online algorithm. This letter indicates a feasibility of sacrificing memory for efficiency in EM computation using an offline–online manner, which has a similar concept with machine learning.

The high regression and generalization ability of deep learning facilitates its application in computational electromagnetics (EMs) [7], [8]. For the EM modeling, deep neural networks have been applied to solve the Poisson's equation [9] and volume integral equation for dielectric targets [10], and approximate boundary conditions [11], [12], etc. Deep learning has also been exploited to solve EM inverse problems [13]–[18]. As far as we know, there are few works solving surface integral equations for conducting targets with the deep learning technique.

In this letter, we propose to use an iterative deep neural network [19] to learn the manner of solving CFIE. Inspired by the conjugate gradient method [20], a quasi-conjugate direction and the update of solution are predicted in each iteration. In the iterative manner, the feedback of data fitness enables the updated solutions to gradually converge toward correct solutions. Less iterations are needed compared with the conventional iterative equation solver. The numerical results validate this method and show the potential of solving the CFIE with the deep learning approach.

II. FORMULATION

In this section, we first review the CFIE for 2-D conducting objects in open region, and then introduce the learning algorithm for predicting the surface current.

A. Combined Field Integral Equation

Consider wave scattering by an infinitely long PEC target, which extends along the $\hat{z}$ -axis. The target boundary on cross section is described by Γ , with its normal direction $\hat{n}$ outward

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Rui Guo, Zhichao Lin, Tao Shan, Maokun Li, Fan Yang, and Shenheng Xu are with the State Key Laboratory on Microwave and Digital Communications, Beijing National Research Center for Information Science and Technology, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: guor17@mails.tsinghua.edu.cn; lzc19@mails.tsinghua.edu.cn; shant16@mails.tsinghua.edu.cn; maokunli@tsinghua.edu.cn; fan_yang@tsinghua.edu.cn; shxu@tsinghua.edu.cn).

Aria Abubakar is with the Mathematic and Modeling, Schlumberger-Doll Research, Houston, TX 02139 USA (e-mail: aabubakar@slb.com).

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and its tangential direction specified as $\hat{\mathbf{t}} = \hat{\mathbf{z}} \times \hat{\mathbf{n}}$. The position vector on the section is specified by $\mathbf{r}$.

For TM_z polarization, assuming the incident electric field $\mathbf{E}^i(\mathbf{r}) = E_z^i(\mathbf{r})\hat{\mathbf{z}}$, an electric current $\mathbf{J}(\mathbf{r}) = J_z(\mathbf{r})\hat{\mathbf{z}}$ is induced on Γ ; J_z satisfies the following EFIE and MFIE [3], [21]:

$$\text{EFIE} : \hat{\mathbf{t}} \frac{k_0}{4} \int_{\Gamma} \bar{J}_z(\mathbf{r}) G_0(\mathbf{r}, \mathbf{r}') d\mathbf{r}' = \hat{\mathbf{t}} E_z^i(\mathbf{r}) \quad (1)$$

$$\text{MFIE} : \hat{\mathbf{z}} \frac{1}{2} \bar{J}_z(\mathbf{r}) + \hat{\mathbf{z}} \int_{\Gamma} \nabla G_0(\mathbf{r}, \mathbf{r}') \bar{J}_z(\mathbf{r}') d\mathbf{r}' = \hat{\mathbf{z}} \bar{H}_t^i(\mathbf{r}) \quad (2)$$

with

$$\begin{aligned} \bar{\mathbf{J}}(\mathbf{r}) &= Z_0 \mathbf{J}(\mathbf{r}), \quad \bar{H}_t^i(\mathbf{r}) = Z_0 \mathbf{H}^i(\mathbf{r}) \\ H_t^i(\mathbf{r}) &= \hat{\mathbf{t}}(-\nabla \times \mathbf{E}^i(\mathbf{r})/j\omega\mu_0) \end{aligned} \quad (3)$$

where k_0 is the wavenumber, $Z_0 = \sqrt{\mu_0/\varepsilon_0}$, and $G_0(\mathbf{r})$ is the 2-D free-space Green's function, i.e.,

$$G_0(\mathbf{r} - \mathbf{r}') = -\frac{j}{4} H_0^{(2)}(k_0 |\mathbf{r} - \mathbf{r}'|). \quad (4)$$

After dividing Γ into small segments and approximating the surface current density over each segment as a constant, two matrix equations for EFIE and MFIE can be established, i.e.,

$$\text{EFIE} : \mathbf{Z}_e \bar{\mathbf{J}} = \mathbf{E} \quad (5)$$

$$\text{MFIE} : \mathbf{Z}_m \bar{\mathbf{J}} = \bar{\mathbf{H}}. \quad (6)$$

The CFIE is in the form of [21]

$$\begin{aligned} \text{CFIE} : \quad & \hat{\mathbf{t}} \frac{k_0}{4} \int_{\Gamma} \bar{J}_z(\mathbf{r}) G_0(\mathbf{r}, \mathbf{r}') d\mathbf{r}' \\ & + \hat{\mathbf{z}} \left(\hat{\mathbf{z}} \frac{1}{2} \bar{J}_z(\mathbf{r}) + \hat{\mathbf{z}} \int_{\Gamma} \nabla G_0(\mathbf{r}, \mathbf{r}') \bar{J}_z(\mathbf{r}') d\mathbf{r}' \right) \\ & = \hat{\mathbf{t}} E_z^i(\mathbf{r}) + \hat{\mathbf{z}} \bar{H}_t^i(\mathbf{r}). \end{aligned} \quad (7)$$

As a result, the matrix equation can be written as

$$(\mathbf{Z}_e + \mathbf{Z}_m) \bar{\mathbf{J}} = \mathbf{E} + \bar{\mathbf{H}}. \quad (8)$$

In the following, we rewrite (8) as

$$\mathbf{A} \bar{\mathbf{J}} = \mathbf{b} \quad (9)$$

for compactness, with $\mathbf{A} = \mathbf{Z}_e + \mathbf{Z}_m$ and $\mathbf{b} = \mathbf{E} + \bar{\mathbf{H}}$.

B. Deep Learning Scheme

Equation (9) can be solved by the iterative method, such as the conjugate gradient (CG) method. In CG, the solution of $\bar{\mathbf{J}}$ in the k th iteration is updated by the following:

- 1) $\mathbf{r}_k = \mathbf{b} - \mathbf{A} \bar{\mathbf{J}}_k$;
- 2) $\beta_{k+1} = (\mathbf{r}_k^T \mathbf{r}_k) / (\mathbf{r}_{k-1}^T \mathbf{r}_{k-1})$;
- 3) $\mathbf{p}_{k+1} = \mathbf{r}_k + \beta_{k+1} \mathbf{p}_k$;
- 4) $\alpha_{k+1} = (\mathbf{r}_k^T \mathbf{r}_k) / \mathbf{p}_{k+1}^T (\mathbf{A} \mathbf{p}_{k+1})$;
- 5) $\bar{\mathbf{J}}_{k+1} = \bar{\mathbf{J}}_k + \alpha_{k+1} \bar{\mathbf{p}}_{k+1}$.

It can be noticed that the conjugate direction $\mathbf{p}_{k+1}$ is computed from $\mathbf{p}_k$, $\mathbf{r}_k$, and $\mathbf{r}_{k-1}$, and the update $\Delta \bar{\mathbf{J}}_{k+1} = \alpha_{k+1} \bar{\mathbf{p}}_{k+1}$ is computed from $\bar{\mathbf{r}}_k$, $\bar{\mathbf{p}}_{k+1}$, and $\bar{\mathbf{A}} \bar{\mathbf{p}}_{k+1}$. For problems with the number of unknowns being N , it could take as

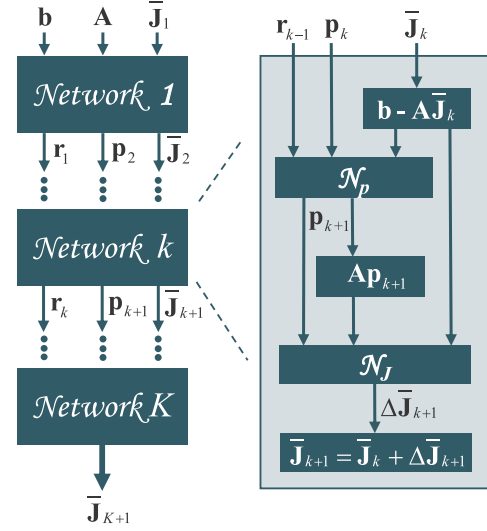


Fig. 1. Diagram of the iterative neural network.

much as N steps to find the accurate solution. As a result, solving (9) for large-scale problems is time consuming.

The high nonlinear regression capability of the deep neural network shows a good potential of accelerating convergence. It can be trained to predict both $\mathbf{p}_k$ and $\Delta \bar{\mathbf{J}}_k$ in an iterative manner. In the k th iteration, $\bar{\mathbf{J}}$ can be updated with predicted intermediates:

- 1) $\mathbf{r}_k = \mathbf{b} - \mathbf{A} \bar{\mathbf{J}}_k$;
- 2) $\mathbf{p}_{k+1} = \mathcal{N}_p^k(\mathbf{p}_k, \mathbf{r}_k, \mathbf{r}_{k-1})$;
- 3) $\bar{\mathbf{J}}_{k+1} = \bar{\mathbf{J}}_k + \mathcal{N}_J^k(\mathbf{p}_{k+1}, \mathbf{A} \mathbf{p}_{k+1}, \mathbf{r}_k)$;

where $\mathcal{N}_p^k$ and $\mathcal{N}_J^k$ are the neural networks in the k th iteration. The diagram of the neural network is shown in Fig. 1.

In the training stage, we adopt supervised learning, i.e., the ground truths of surface current density are labels. In addition, the input of the neural network is the $\mathbf{A}$ matrix established by MoM. Several subnetworks sharing the same structure are cascaded, with each subnetwork representing one iteration for updating the current. The matrix-vector products, $\mathbf{A} \bar{\mathbf{J}}$ and $\mathbf{A} \mathbf{p}$, are computed strictly in each subnetwork. It is worth noting that compared with the noniterative scheme, the iterative manner provides the network more opportunities to adjust the predicted results according to the feedback from data fitness.

III. EXPERIMENTAL RESULTS

In the following, numerical examples of solving CFIE are presented. The training dataset is generated by MoM written in Python 3.7. The deep neural network is designed using the Tensorflow library. The computation platform is with one 48-core Intel(R) Xeon(R) Gold 5118 CPU 2.3 GHz, and one Nvidia(R) Tesla(R) V100 GPU.

A. Experimental Setup

With reference to Fig. 2, the incident plane wave is along $+x$ -direction, with its frequency being 3 GHz. The conducting target is described in polar coordinate, and the radial coordinate varies

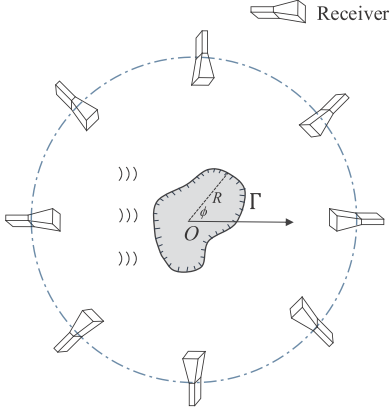


Fig. 2. Experimental setup for the scattering problem.

according to

$$R = A_0 + A_1 \sin(\omega_1 \phi + \theta_1) + A_2 \sin(\omega_2 \phi + \theta_2). \quad (10)$$

To generate training and testing dataset, we randomly select $A_0 \in (0.8, 1)$, $A_1 \in (0.1, 0.3)$, $A_2 \in (0.2, 0.4)$, $\omega_1 \in (0, 8)$, $\omega_2 \in (1, 6)$, and $\theta_1, \theta_2 \in (0, \pi)$. Therefore, the diameter of the object is in the order of $10\text{--}20\lambda$. For numerical computation, the surface Γ is divided into 720 segments. Once the surface current density is obtained, with the scattered E-field at the receivers location computed by

$$E_z^s(\mathbf{r}^R) = -jk_0 Z_0 \oint_{\Gamma} G_0(\mathbf{r}^R - \mathbf{r}') J_z(\mathbf{r}') d\mathbf{r}' \quad (11)$$

the bistatic scattering cross section can be obtained according to

$$\sigma_{2D} = 10 \log_{10} \left(2\pi r \left| \frac{E_z^s}{E_z^i} \right|^2 \right). \quad (12)$$

In numerical simulations, we benchmark the network performance using both the current density and the computed bistatic scattering cross section.

We design the $\mathcal{N}_p$ and $\mathcal{N}_J$ using the fully connected neural network. The number of hidden layers in $\mathcal{N}_p$ and $\mathcal{N}_J$ is four, with 180 nodes contained in each layer. Dropout layers with the rate 0.2 are added after every hidden layer to eliminate the overfitting in training. The number of iterative subnetworks is $K = 8$. Totally, there are 20 000 pairs of $\mathbf{A}$ and $\bar{\mathbf{J}}$ for training, and 1000 pairs for testing. The training stage is done using the ADAM method [22] with the learning rate starting at 10^{-4} and decaying exponentially.

B. Numerical Tests

For the 1000 testing cases, we compute the normalized misfit between predicted current density and the ground truth according to

$$\text{rms} = \frac{\|\mathbf{J}^{\text{true}} - \mathbf{J}^{\text{pred}}\|_2}{\|\mathbf{J}^{\text{true}}\|_2} \quad (13)$$

where $\mathbf{J}^{\text{true}}$ denotes the true current density and $\mathbf{J}^{\text{pred}}$ denotes the predicted current density. The histogram of the normalized misfit is shown in Fig. 3, where the mean and standard deviation

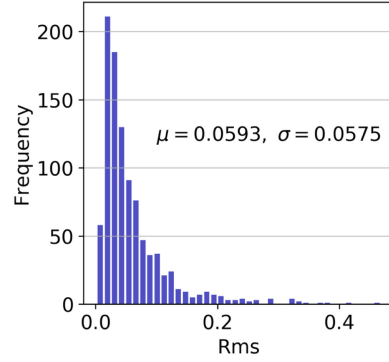


Fig. 3. Statistical histograms of the normalized misfit between the predicted surface current density and the ground truth. A total of 1000 testing targets are tested.

is 0.059 and 0.057, respectively. The statistics show that the error of the network remains in an acceptable level.

We select three of the 1000 cases and present the corresponding surface currents and bistatic scattering cross sections σ_{2D} in Fig. 4, namely Test #1, #2, and #3. The predicted currents and the σ_{2D} computed from predicted currents are plotted with blue dotted lines, while the ground truths are plotted in yellow lines. Although the true currents have large oscillations, the network can make prediction with good accuracy. In addition, the computed bistatic scattering cross sections agree well with the ground truth.

Furthermore, the target with nonconvex outline is tested, although the training set only contains convex polygons. The result is shown in Fig. 4 Test #4. Although the outline of the testing one and training models differs significantly, the predicted current and the σ_{2D} still have a good match with the ground truth. This example partly indicates the generalization ability of this network. Further improvements can be made by considering more nonconvex polygons in the training set.

In the testing stage, the number of iterations for the subnetworks is 8. For comparison, we use the BiCGSTAB [23] method to solve the matrix equations of the testing cases. The average number of iterations with BiCGSTAB is 34.

Once the $\mathbf{A}$ matrix is established, the average prediction time for each case is less than 0.005 s. However, we notice that using BiCGSTAB can achieve a similar or slightly faster speed compared with the neural network. This is mainly because for 2-D case, the number of unknowns is relatively small. But when the problem scale is large, this learning-based scheme has the potential of accelerating conventional iterative solver by reducing the number of matrix–vector products.

IV. DISCUSSION

We provide a supervised learning scheme to train the neural network, i.e., the loss function is defined as $\text{loss} = \|\bar{\mathbf{J}} - \bar{\mathbf{J}}^*\|_2^2$, where the $\bar{\mathbf{J}}$ and $\bar{\mathbf{J}}^*$ denote the output and label, respectively. But theoretically, it is possible to train the neural network in self-supervised manner, such as $\text{loss} = \|\mathbf{b} - \mathbf{A}\bar{\mathbf{J}}\|_p^p$ with $\|\cdot\|_p$ being the L_p -norm. We tried $p=1$ and 2, but the training loss is difficult to converge to a level for predicting currents accurately.

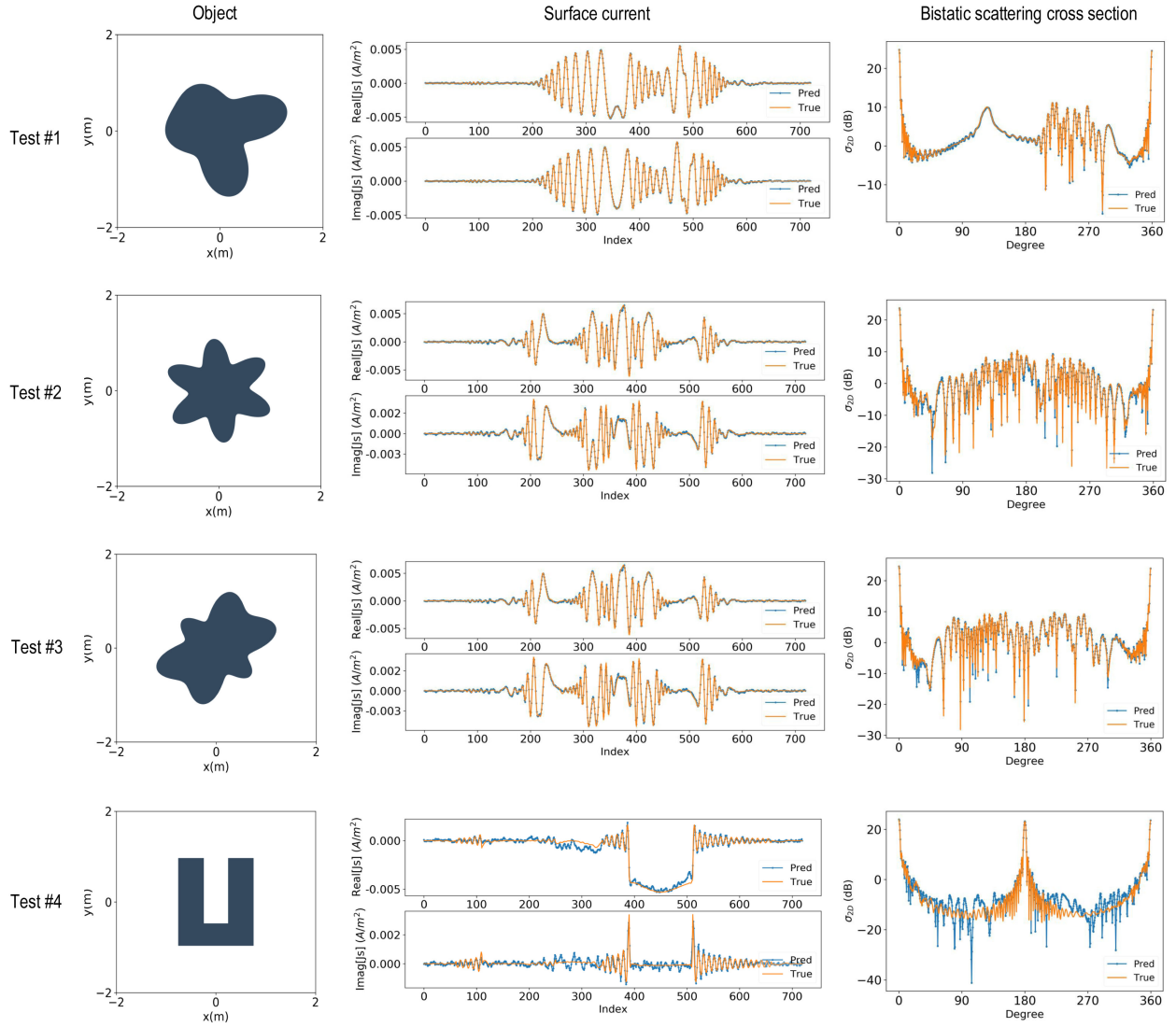


Fig. 4. Four samples of the testing cases. The first column shows the 2-D cross section of the conducting objects. The second column shows the real and imaginary parts of surface current density, with the blue and yellow lines representing the predicted and true current, respectively. The third column shows the bistatic scattering cross sections, with the blue and yellow lines representing the ones computed from the predicted and true current, respectively.

We also added the regularization using structural similarity index measure between $\mathbf{b}$ and $\mathbf{A}\bar{\mathbf{J}}$, but there is no significant improvement.

Second, when generate training objects, it is important to describe the surface with fixed number of unknowns in predefined sequence. To deal with scale-independent problems, the fully convolutional neural network is a choice, but this will make the incident field only has local impact on surface current. In this case, the kernel size should be carefully designed. Another solution may be zero-padding that fills the corresponding elements in $\mathbf{A}$ matrix and vector with zeros for those nodes not on the surface. Besides, the targets are parameterized in polar coordinates, but more flexible manner should be applied to represent different types of shapes.

More advanced techniques in the neural network design can also further improve the performance. We will keep on studying the aforementioned problems and refining the

neural network. We hope it can be extended to 3-D case in the future.

V. CONCLUSION

In this letter, we apply an iterative deep neural network to solve the CFIE for 2-D conducting objects. The input is the coefficient matrix of the CFIE while the output is the surface current density. Both physical information and historical knowledge of solving the CFIE are utilized to predict the surface current. The nonlinearity of the neural network can accelerate the convergence of the iterative solver with good accuracy. In an iterative manner, the feedback of data fitness enables the predicted current to gradually converge toward correct solutions. The feasibility and generalization ability are validated by numerical examples. This method shows a new perspective of solving surface integral equation with the deep learning technique.

REFERENCES

- [1] W. C. Chew, *Waves and Fields in Inhomogeneous Media*. New York, NY, USA: IEEE Press, 1995.
- [2] X. Chen, *Computational Methods for Electromagnetic Inverse Scattering*. Hoboken, NJ, USA: Wiley, 2018.
- [3] J.-M. Jin, *Theory and Computation of Electromagnetic Fields*. Hoboken, NJ, USA: Wiley, 2011.
- [4] W. C. Chew, E. Michielssen, J. Song, and J.-M. Jin, *Fast and Efficient Algorithms in Computational Electromagnetics*. Norwood, MA, USA: Artech House, 2001.
- [5] M. Ganesh, J. S. Hesthaven, and B. Stamm, "A reduced basis method for electromagnetic scattering by multiple particles in three dimensions," *J. Comput. Phys.*, vol. 231, no. 23, pp. 7756–7779, 2012.
- [6] X. Dang, M. Li, F. Yang, and S. Xu, "Quasi-periodic array modeling using reduced basis method," *IEEE Antennas Wireless Propag. Lett.*, vol. 16, pp. 825–828, 2017.
- [7] X. Chen, Z. Wei, M. Li, and P. Rocca, "A review of deep learning approaches for inverse scattering problems," *Prog. Electromagn. Res.*, vol. 167, pp. 67–81, 2020.
- [8] A. Massa, G. Oliveri, M. Salucci, N. Anselmi, and P. Rocca, "Learning-by-examples techniques as applied to electromagnetics," *J. Electromagn. Waves Appl.*, vol. 32, no. 4, pp. 516–541, 2018.
- [9] T. Shan *et al.*, "Study on a fast solver for Poisson's equation based on deep learning technique," *IEEE Trans. Antennas Propag.*, vol. 68, no. 9, pp. 6725–6733, Sep. 2020.
- [10] S. Qi, Y. Wang, Y. Li, X. Wu, Q. Ren, and Y. Ren, "Two-dimensional electromagnetic solver based on deep learning technique," *IEEE J. Multiscale Multiphys. Comput. Tech.*, vol. 5, pp. 83–88, 2020.
- [11] H. M. Yao and L. Jiang, "Machine-learning-based PML for the FDTD method," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, no. 1, pp. 192–196, Jan. 2019.
- [12] H. M. Yao and L. Jiang, "Enhanced PML based on the long short term memory network for the FDTD method," *IEEE Access*, vol. 8, pp. 21 028–21 035, 2020.
- [13] Y. Jin, Q. Shen, X. Wu, J. Chen, and Y. Huang, "A physics-driven deep-learning network for solving nonlinear inverse problems," *Petrophysics*, vol. 61, no. 1, pp. 86–98, 2020.
- [14] L. Li, L. G. Wang, F. L. Teixeira, C. Liu, A. Nehorai, and T. J. Cui, "DeepNIS: Deep neural network for nonlinear electromagnetic inverse scattering," *IEEE Trans. Antennas Propag.*, vol. 67, no. 3, pp. 1819–1825, Mar. 2019.
- [15] Z. Wei and X. Chen, "Deep-learning schemes for full-wave nonlinear inverse scattering problems," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 1849–1860, Apr. 2019.
- [16] R. Guo *et al.*, "Pixel-and model-based microwave inversion with supervised descent method for dielectric targets," *IEEE Trans. Antennas Propag.*, vol. 68, no. 12, pp. 8114–8126, Dec. 2020.
- [17] Z. Lin *et al.*, "Neural network based supervised descent method for 2D electrical impedance tomography," *Physiol. Meas.*, vol. 41, no. 7, 2020, Art. no. 074003.
- [18] X. Ye, Y. Bai, R. Song, K. Xu, and J. An, "An inhomogeneous background imaging method based on generative adversarial network," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 11, pp. 4684–4693, Nov. 2020.
- [19] J. Adler and O. Öktem, "Solving ill-posed inverse problems using iterative deep neural networks," *Inverse Problems*, vol. 33, no. 12, 2017, Art. no. 124007.
- [20] M. R. Hestenes and E. Stiefel, "Methods of conjugate gradients for solving linear systems," *J. Res. Nat. Bur. Standards*, vol. 49, no. 6, pp. 409–436, 1952.
- [21] F. P. Andriulli and E. Michielssen, "A regularized combined field integral equation for scattering from 2-D perfect electrically conducting objects," *IEEE Trans. Antennas Propag.*, vol. 55, no. 9, pp. 2522–2529, Sep. 2007.
- [22] D. P. Kingma and J. Ba, "ADAM: A method for stochastic optimization," *Proc. 3rd Int. Conf. Learn. Representations (ICLR)*, 2014.
- [23] H. A. Van der Vorst, "Bi-CGSTAB: A fast and smoothly converging variant of Bi-CG for the solution of nonsymmetric linear systems," *SIAM J. Sci. Statist. Comput.*, vol. 13, no. 2, pp. 631–644, 1992.